HDOC Day 37
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Batch - 15
Algorithm

Main Program

1. Start.
2. Display the menu.
3. Read the user's choice.
4. Use `switch` to call the corresponding function.
5. For choice 1, call `toUppercase()`.
6. For choice 2, call `reverseString()`.
7. For choice 3, call `checkPalindrome()`.
8. For choice 4, exit.
9. For any other choice, display an error message.
10. Repeat until choice 4 is selected.
11. Stop.

Operation 1 – Lowercase to Uppercase

1. Read the string.
2. Traverse characters until `'\0'`.
3. If a character lies between `'a'` and `'z'`, convert it using:

```
str[i] = str[i] - ('a' - 'A')
```

4. Display the uppercase string.

Operation 2 – Reverse String

1. Read the string.
2. Determine its length by traversing until `'\0'`.
3. Swap the first character with the last, second with second-last, and so on.
4. Continue until half of the string has been processed.
5. Display the reversed string.

Operation 3 – Palindrome

1. Read the string.
2. 3. 4. 5. Determine its length without `strlen()`.

Compare the first character with the last, second with second-last, etc.
If any pair differs, the string is not a palindrome.
Otherwise, it is a palindrome.

Test Case Table

Test	Choice	Input	Expected Output
1	1	hello world	HELLO WORLD
2	1	computer	COMPUTER
3	2.	hello	olleh
4	2	programming	gnimmargorp
5	3.	madam.	Palindrome
6	3	level.	Palindrome
7	3	hello	Not palindrome
8	5 — Invalid choice		
9	4 — Exit		

C Program

```
#include <stdio.h>
void toUppercase(void);
void reverseString(void);
void checkPalindrome(void);
int main()
{
int choice;
do
{
printf("\n===== STRING OPERATIONS =====\n");
printf("1. Convert lowercase string to uppercase\n");
printf("2. Reverse a string\n");
printf("3. Check if a string is a palindrome\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
toUppercase();
break;
case 2:
reverseString();
break;
case 3:
checkPalindrome();
break;
```

```
case 4:
printf("Exiting program...\n");
break;
default:
printf("Invalid choice! Please enter 1 to 4.\n");
}
} while (choice != 4);
return 0;
}
/* Function to convert lowercase string to uppercase */
void toUppercase(void)
{
char str[100];
int i;
printf("Enter a lowercase string: ");
scanf(" %[^\n]", str);
for (i = 0; str[i] != '\0'; i++)
{
if (str[i] >= 'a' && str[i] <= 'z')
{
str[i] = str[i] - ('a' - 'A');
}
}
printf("Uppercase string: %s\n", str);
}
/* Function to reverse a string */
void reverseString(void)
{
char str[100], temp;
int i, length = 0;
printf("Enter a string: ");
scanf(" %[^\n]", str);
/* Find length without strlen() */
while (str[length] != '\0')
{
length++;
}
{
for (i = 0; i < length / 2; i++)
}
temp = str[i];
str[i] = str[length - 1 - i];
str[length - 1 - i] = temp;
```

```
printf("Reversed string: %s\n", str);
}
/* Function to check palindrome */
void checkPalindrome(void)
{
char str[100];
int i, length = 0;
int palindrome = 1;
printf("Enter a string: ");
scanf(" %[^\n]", str);
/* Find length without strlen() */
while (str[length] != '\0')
{
length++;
}
{
for (i = 0; i < length / 2; i++)
if (str[i] != str[length - 1 - i])
{
palindrome = 0;
break;
}
}
if (palindrome)
printf("The string is a palindrome.\n");
else
printf("The string is not a palindrome.\n");
```